

## **Part 1 - Software Project Planning**

### **Project Description**

The **Campus Event Tracker** is a web-based application designed to provide students with centralized access to upcoming campus events. It will display event information such as schedules, categories, and locations, allowing users to easily browse and search for activities of interest. The system aims to enhance student involvement by offering a convenient and organized event discovery platform.

### **Project Scope**

#### **Functional Requirements**

1. The system shall display a list of upcoming campus events with essential details.
2. The system shall allow users to filter or search events based on criteria such as date, category, or department.
3. The system shall provide detailed information for each event, including time, location, description, and contact information.

#### **Task Breakdown**

1. **Project Initialization** – Set up the Angular project structure, dependencies, and environment configuration.

2. **Event List Component Development** – Implement the component responsible for rendering and managing the list of events.
3. **Event Service Implementation** – Create the service layer to handle event data retrieval, storage, and processing.
4. **User Interface Layout and Styling** – Design and develop the visual layout for the homepage, event cards, and navigation elements.
5. **Testing and Input Validation** – Conduct functional testing and ensure proper handling of user input and edge cases.

## **Estimation**

### **Create Database Schema**

- Estimation: 5 (Medium complexity)
- This task involves designing tables and relationships for events and users, which is crucial but not highly complex.

### **Create User Authentication Service**

- Estimation: 8 (High complexity)
- Building secure user authentication, including password reset functionality, requires attention to security and handling edge cases.

### **Design Event Management UI**

- Estimation: 5 (Medium complexity)
- The UI design will require some effort, but it's more about layout and integration with backend services.

## **Implement Event RSVP Functionality**

- Estimation: 5 (Medium complexity)
- The RSVP feature requires integrating event data and user actions, but it's straightforward after the event management functionality is in place.

## **Test Application Features**

- Estimation: 3 (Low complexity)
- Testing the app involves ensuring that all user stories and features work as expected, and that bugs are identified and resolved.

## **Part 2 - Software Architecture Design**

### **Chosen Architecture Style**

Component-Based Architecture

### **Architecture Explanation**

In Angular, the Component-Based Architecture organizes the application into modular, self-contained components, each responsible for a specific UI section and its related logic. Components communicate through inputs, outputs, and shared services, promoting separation of concerns and consistent data flow. This architecture also ensures that updates to one component do not negatively impact others, supporting maintainability and scalable development.

### **Justification**

The Component-Based Architecture is highly suitable for the Campus Event Tracker because it enables the application to be developed as a collection of independent, reusable components such as event listings, filters, and detail views. This modularity reduces complexity, simplifies maintenance, and allows for rapid addition of new features as the system grows. Its structured and organized approach makes it ideal for small to medium-sized applications requiring clarity, flexibility, and scalability.